



STUDENT PERFORMANCE ANALYSIS USING SQL AND PYTHON

A Data Analysis Project

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Project Focus:

Analysis of student academic performance using SQL, Python, Pandas, and Matplotlib to identify factors associated with final grades and classify students according to their academic performance.

Introduction

Student performance is influenced by several academic and behavioural factors, including previous academic results, study time, failures, attendance, and other student-related characteristics. Analysing these factors can help identify patterns in academic performance and highlight students who may require additional academic support.

This project focuses on analysing a student performance dataset containing **395 student records and 33 variables**. The analysis was performed using **SQL and Python**. SQL was used to perform structured queries, aggregation, grouping, and comparison of student data, while Python, particularly **Pandas and Matplotlib**, was used for data cleaning, statistical analysis, visualization, and interpretation.

The main objective of the project is not only to calculate student grades, but to identify the factors that have the strongest relationship with the final grade (**G3**) and to divide students into meaningful performance groups.

Objectives of the Project

The major objectives of this project are:

1. To understand the structure and quality of the student performance dataset.
2. To calculate the overall average, highest, and lowest final grades.
3. To compare academic performance between different groups of students.
4. To investigate the relationship between study time and final grades.
5. To examine the effect of previous failures on final academic performance.
6. To analyse the relationship between absences and final grades.
7. To investigate the relationship between previous academic grades and the final grade.
8. To identify students who may be considered academically at risk.
9. To use visualizations to communicate important patterns in the dataset.
10. To demonstrate the practical use of SQL and Python together in a data analytics workflow.

Dataset Overview

The dataset contains information about students and their academic performance.

The dataset contains:

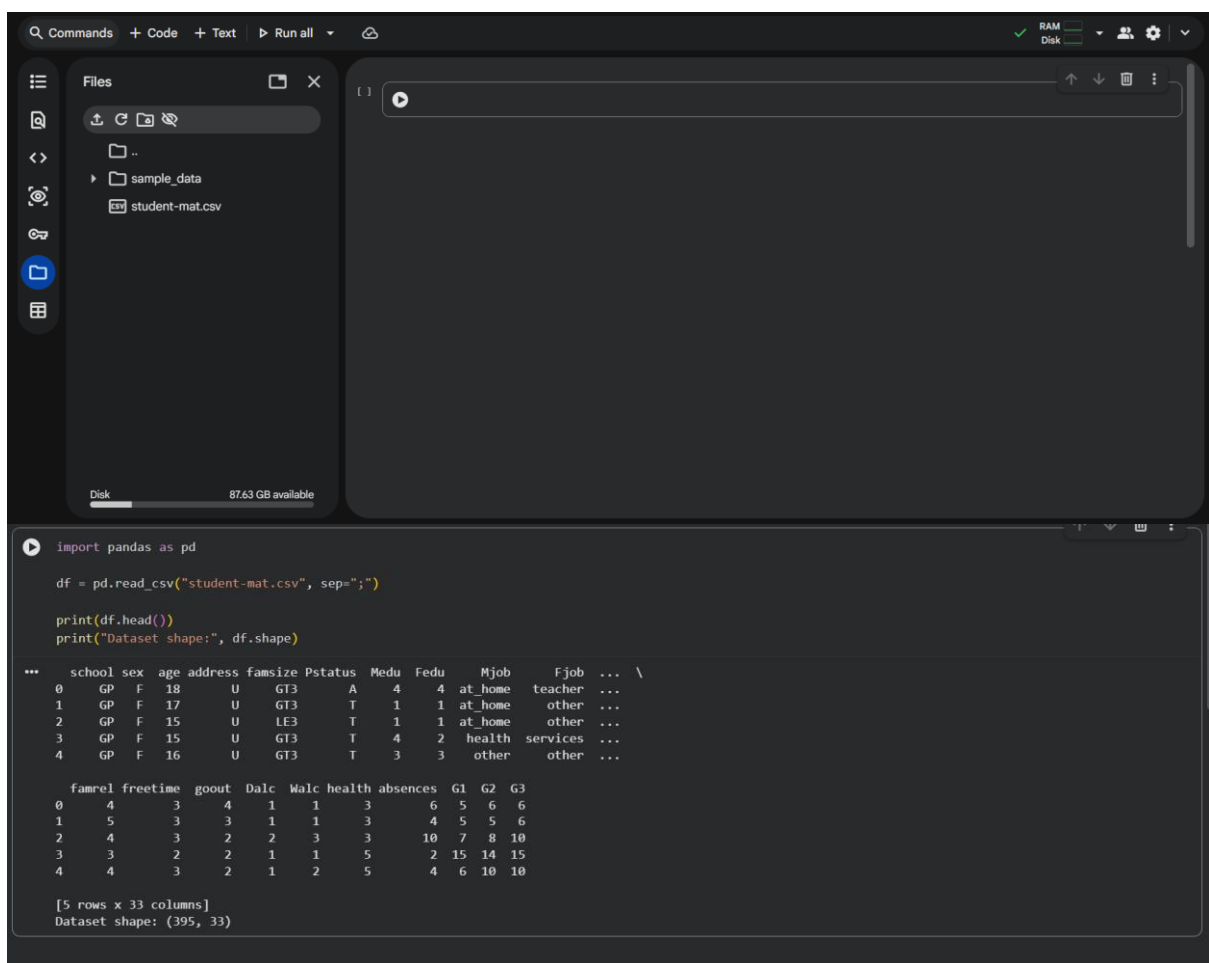
- **395 student records**
- **33 variables**
- Academic information such as **G1, G2 and G3**
- **Study time**
- **Number of previous failures**
- **Absences**

- Gender and school information
- Family and educational background
- Other student-related attributes

The variable **G3** represents the student's final grade and was therefore selected as the primary outcome variable for the analysis.

Loading the Dataset

The dataset was imported into Python using the Pandas library. Since the original CSV file uses a semicolon (;) as the delimiter, the `sep=";"` parameter was used while loading the data.



```
import pandas as pd

df = pd.read_csv("student-mat.csv", sep=";")

print(df.head())
print("Dataset shape:", df.shape)
```

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	...	\
0	GP	F	18	U	GT3	A	4	4	at home	teacher	...	
1	GP	F	17	U	GT3	T	1	1	at home	other	...	
2	GP	F	15	U	LE3	T	1	1	at home	other	...	
3	GP	F	15	U	GT3	T	4	2	health	services	...	
4	GP	F	16	U	GT3	T	3	3	other	other	...	

	famrel	freetime	goout	Dalc	Walc	health	absences	G1	G2	G3
0	4	3	4	1	1	3	6	5	6	6
1	5	3	3	1	1	3	4	5	5	6
2	4	3	2	2	3	3	10	7	8	10
3	3	2	2	1	1	5	2	15	14	15
4	4	3	2	1	2	5	4	6	10	10

[5 rows x 33 columns]
Dataset shape: (395, 33)

Figure 1: Loading the student performance dataset into Python.

The output confirms that the dataset contains **395 rows and 33 columns**, providing sufficient information for analysing different factors associated with student performance.

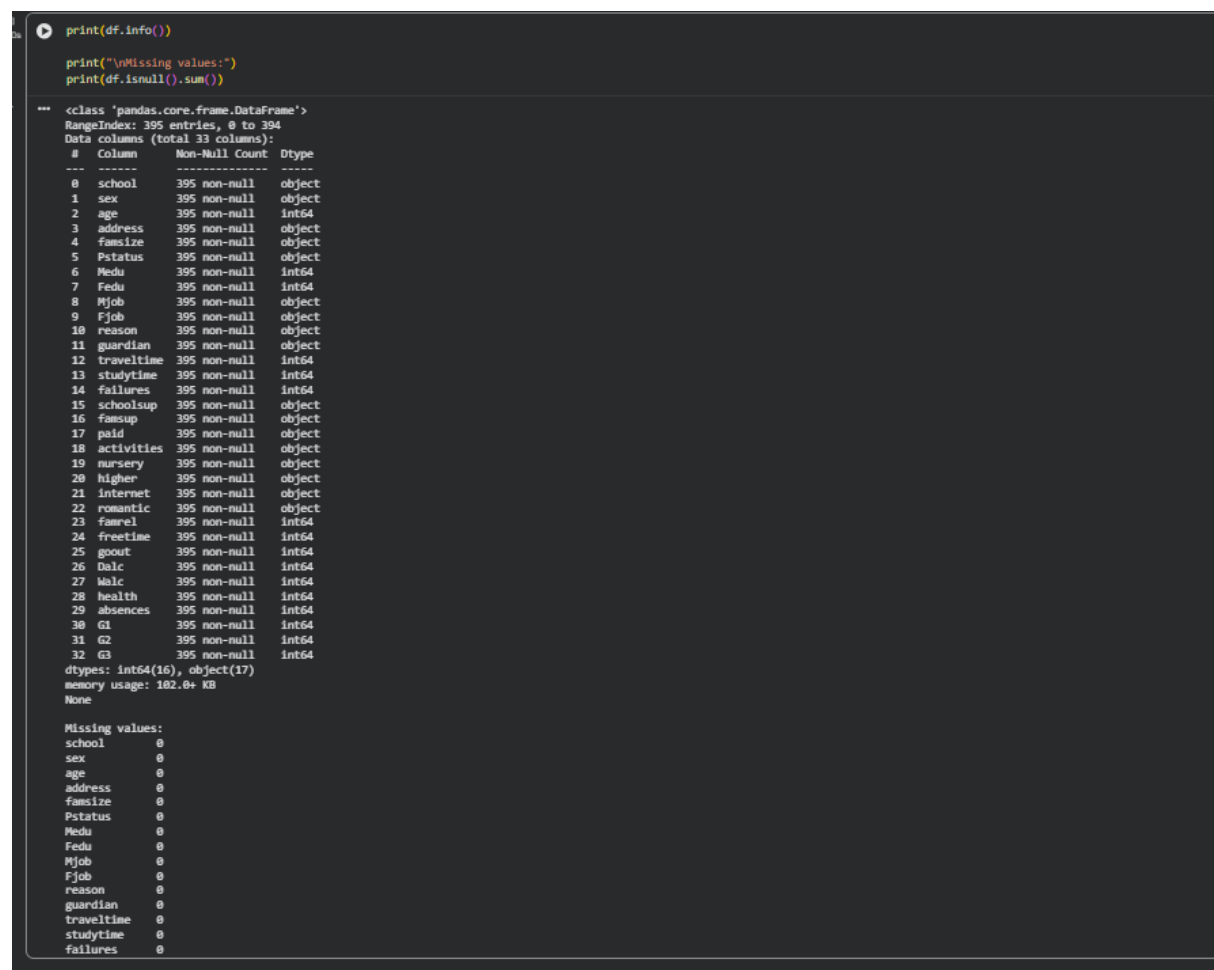
4. DATA QUALITY CHECK

Data Structure and Missing Values

Before performing the analysis, the dataset was inspected to understand its structure, data types, and completeness. The `df.info()` function was used to identify the number of records, variables, and data types.

A missing-value check was also performed using:

```
df.isnull().sum()
```



```
print(df.info())

print("\nMissing values:")
print(df.isnull().sum())

'''
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 395 entries, 0 to 394
Data columns (total 33 columns):
 #   Column              Non-Null Count  Dtype  
---  --
 0   school              395 non-null   object 
 1   sex                 395 non-null   object 
 2   age                 395 non-null   int64  
 3   address             395 non-null   object 
 4   famsize             395 non-null   object 
 5   Pstatus             395 non-null   object 
 6   Medu                395 non-null   int64  
 7   Fedu                395 non-null   int64  
 8   Mjob                 395 non-null   object 
 9   Fjob                 395 non-null   object 
10   reason              395 non-null   object 
11   guardian            395 non-null   object 
12   traveltime          395 non-null   int64  
13   studytime           395 non-null   int64  
14   failures            395 non-null   int64  
15   schoolsup            395 non-null   object 
16   famsup              395 non-null   object 
17   paid                395 non-null   object 
18   activities          395 non-null   object 
19   nursery             395 non-null   object 
20   higher              395 non-null   object 
21   internet            395 non-null   object 
22   romantic            395 non-null   object 
23   famrel              395 non-null   int64  
24   freetime            395 non-null   int64  
25   goout               395 non-null   int64  
26   Dalc                395 non-null   int64  
27   Walc                395 non-null   int64  
28   health              395 non-null   int64  
29   absences            395 non-null   int64  
30   G1                  395 non-null   int64  
31   G2                  395 non-null   int64  
32   G3                  395 non-null   int64  
dtypes: int64(16), object(17)
memory usage: 182.0+ KB
None

Missing values:
school      0
sex         0
age         0
address     0
famsize     0
Pstatus     0
Medu        0
Fedu        0
Mjob        0
Fjob        0
reason      0
guardian    0
traveltime  0
studytime  0
failures    0'''
```

Figure 2: Dataset structure and missing-value analysis.

The analysis showed that all **395 records** contained **valid values across the available variables**, with no missing values detected. This meant that additional missing-value treatment was not required before proceeding with the analysis.

SQL-Based Data Analysis

SQL was used to perform initial exploratory analysis of the student dataset. Aggregate functions such as `COUNT()` and `AVG()` and the `GROUP BY` clause were used to answer questions about the number of students and differences in academic performance between groups.

SQL analysis included:

- Total number of students
- Number of students by gender
- Average final grade by gender
- Average study time and final grade
- Average final grade by school

Student Distribution by Gender

SQL was used to determine the number of students belonging to each gender category. This provides an initial understanding of the composition of the dataset.

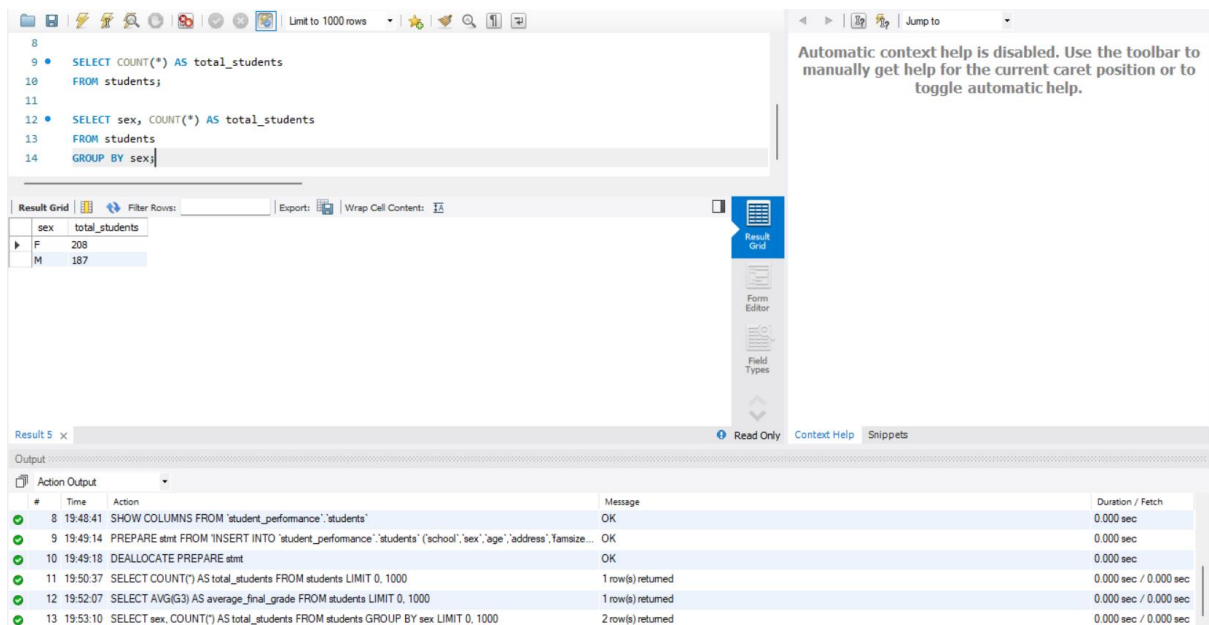


Figure 3: Number of students grouped by gender using SQL.

Average Final Grade by Gender

The average final grade was calculated for each gender using SQL aggregation and grouping.

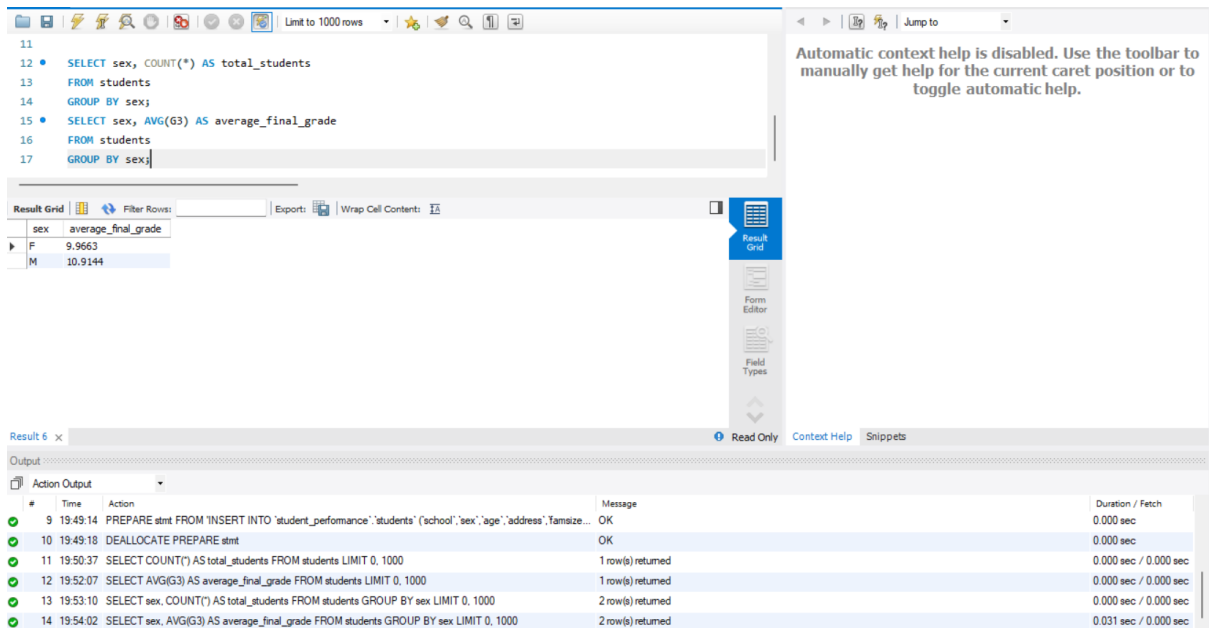


Figure 4: Average final grade by gender.

The results show an average final grade of approximately **9.97 for female students** and **10.91 for male students**. Although a difference is visible, this analysis describes the dataset and does not by itself establish that gender causes differences in academic performance.

Average Final Grade by School

The final grades were grouped according to school to identify whether differences existed between the two schools represented in the dataset.

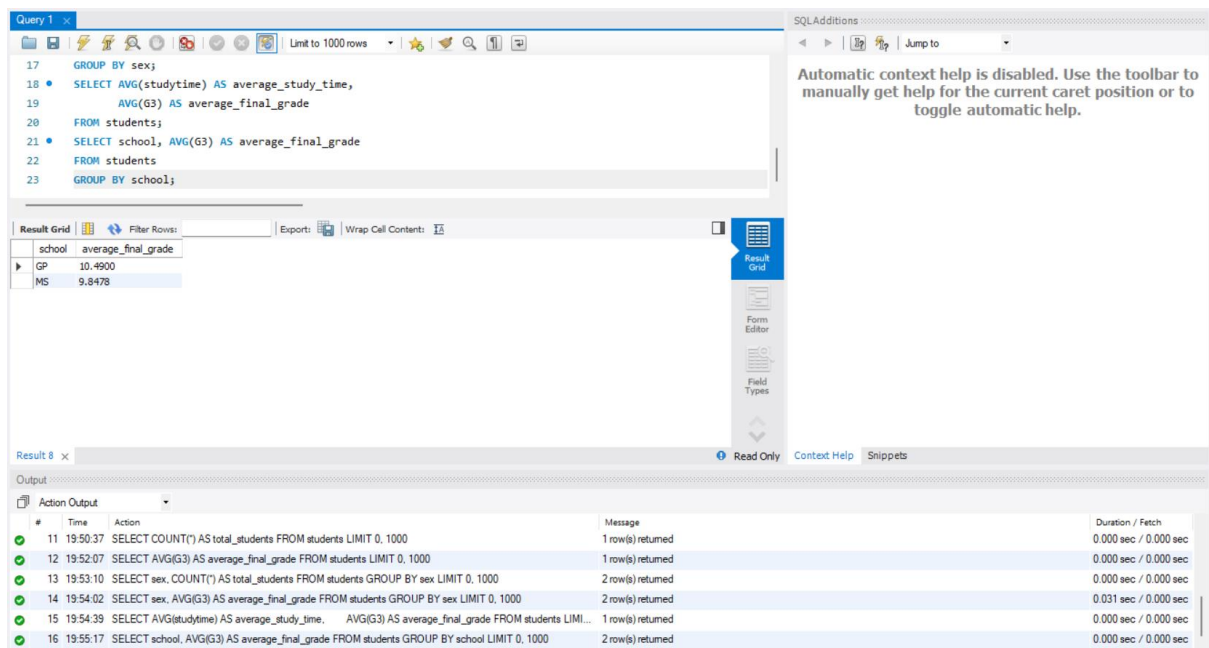


Figure 5: Average final grade by school.

The results show an average final grade of approximately **10.49 for GP students** and **9.85 for MS students**.

Overall Final Grade Analysis

Python was used to calculate basic statistical measures for the final grade variable (**G3**).

The analysis identified:

- **Average final grade:** 10.42
- **Highest final grade:** 20
- **Lowest final grade:** 0

These values provide a general overview of academic performance in the dataset.

DISTRIBUTION OF FINAL GRADES

A histogram was created using Matplotlib to examine how the final grades were distributed among the 395 students.

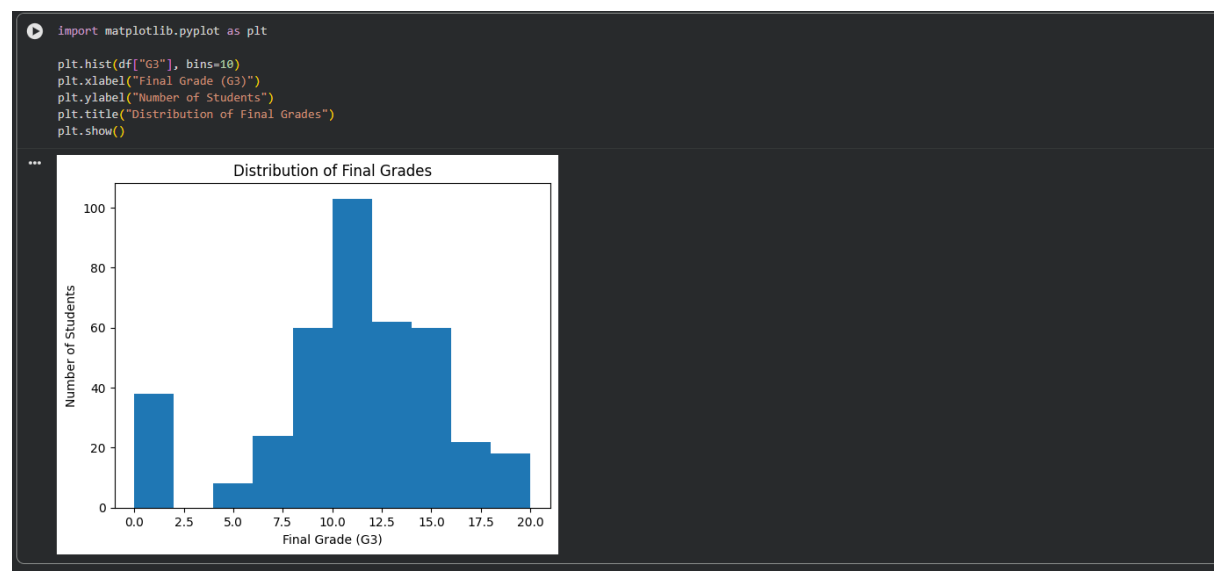


Figure 6: Distribution of final grades (G3).

The distribution shows that a large proportion of students achieved final grades around the middle of the grading scale, while a smaller group achieved very high or very low grades. The presence of students with a final grade of zero is also visible.

This visualization helps demonstrate that student performance is not evenly distributed across the grading scale.

Relationship Between Study Time and Final Grade

The average final grade was calculated for each study-time level to investigate whether increased study time was associated with better academic performance.

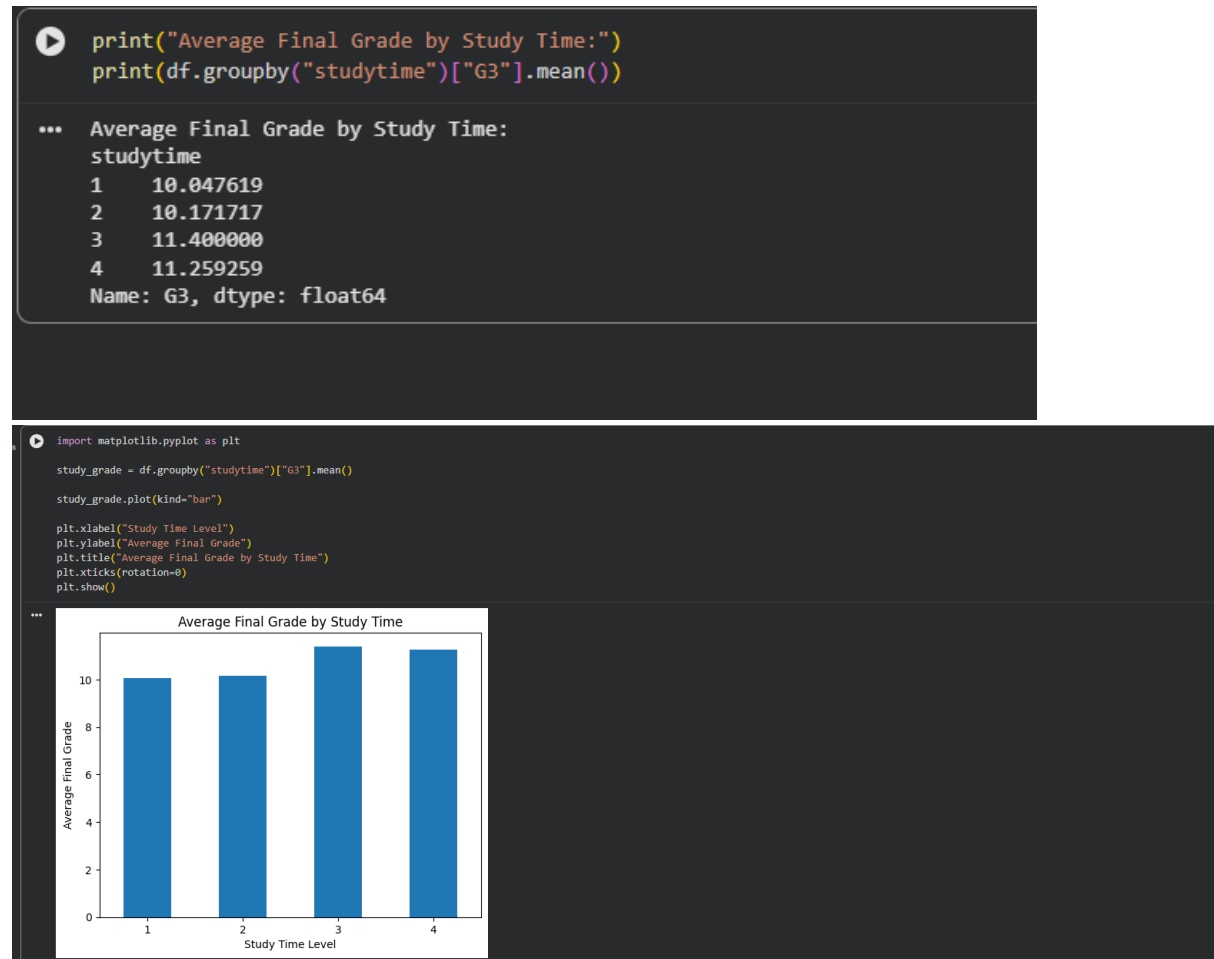


Figure 7: Average final grade by study-time level.

The results were:

Study Time Level Average G3

1	10.05
2	10.17
3	11.40
4	11.26

The results indicate that students in the higher study-time groups generally achieved higher average final grades than students in the lower study-time groups.

However, the correlation between study time and final grade was only **0.0978**, indicating that study time alone has a relatively weak linear relationship with final performance.

Important conclusion:

Study time appears to have some group-level association with performance, but it is not a strong predictor of final grade by itself.

Impact of Previous Failures

The number of previous academic failures was analysed to determine whether students with a history of failures tended to have lower final grades.

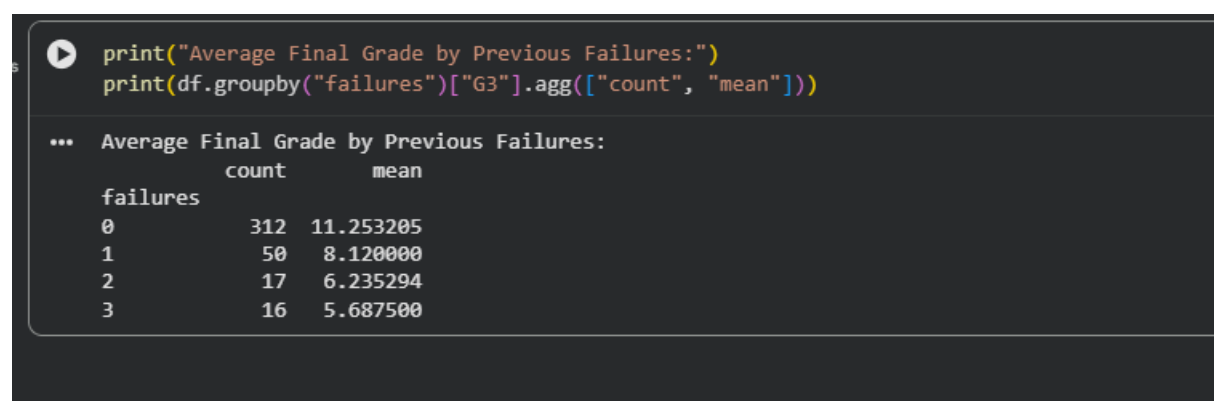


Figure 8: Average final grade by number of previous failures.

Previous Failures Students Average G3

0	312	11.25
1	50	8.12
2	17	6.24
3	16	5.69

A clear downward pattern can be observed. Students with no previous failures had an average final grade of approximately **11.25**, while students with three previous failures had an average of approximately **5.69**.

This represents one of the strongest findings of the project.

Students with a greater history of previous failures tended to have substantially lower final grades.

Analysis of Absence Levels

Absence levels were grouped into four categories to make the analysis easier to interpret:

- 0–5 absences
- 6–10 absences
- 11–20 absences
- 21+ absences

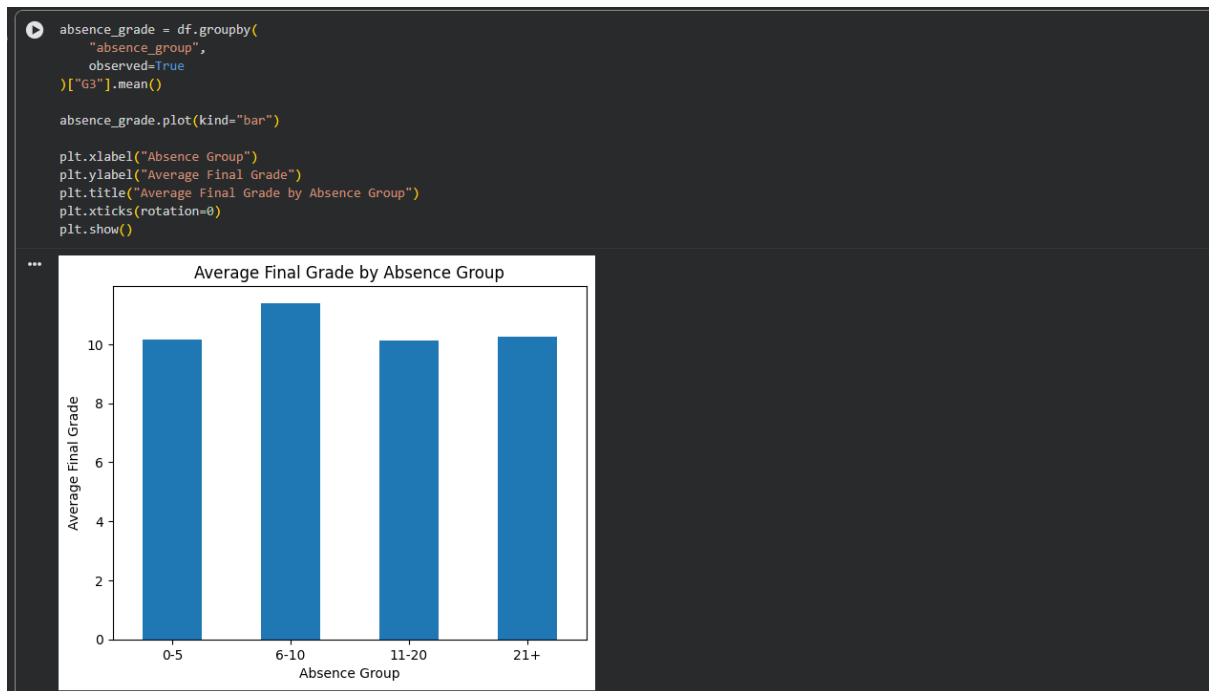


Figure 9: Average final grade by absence group.

Absence Group Average G3

0–5	10.17
6–10	11.40
11–20	10.12
21+	10.27

Interestingly, the results do **not** show a simple relationship where more absences always result in lower grades.

The correlation between absences and final grade was only **0.0342**, which indicates an extremely weak linear relationship in this dataset.

Finding:

Absence count alone was not a strong predictor of final grade in this dataset.

Relationship Between Previous Grades and Final Grade

The relationship between previous academic grades and the final grade was investigated using correlation analysis.

The correlation results were:

Variable	Correlation with G3
G2	0.9049
G1	0.8015
Study Time	0.0978
Absences	0.0342
Failures	-0.3604

```
print("Correlation with Final Grade (G3):")
print(df[["G1", "G2", "studytime", "failures", "absences", "G3"]].corr()["G3"].sort_values(ascending=False))

*** Correlation with Final Grade (G3):
G3      1.000000
G2      0.904868
G1      0.801468
studytime 0.097820
absences  0.034247
failures -0.360415
Name: G3, dtype: float64
```

Figure 10: Correlation of selected variables with final grade.

The results show that **G2 has the strongest positive correlation with the final grade, at approximately 0.905**. G1 also has a strong positive correlation of approximately **0.802**.

Previous academic grades therefore appear to be much more strongly associated with final performance than study time or absence levels.

G2 and Final Grade Relationship

A scatter plot was created to visually examine the relationship between the second-period grade (G2) and final grade (G3).

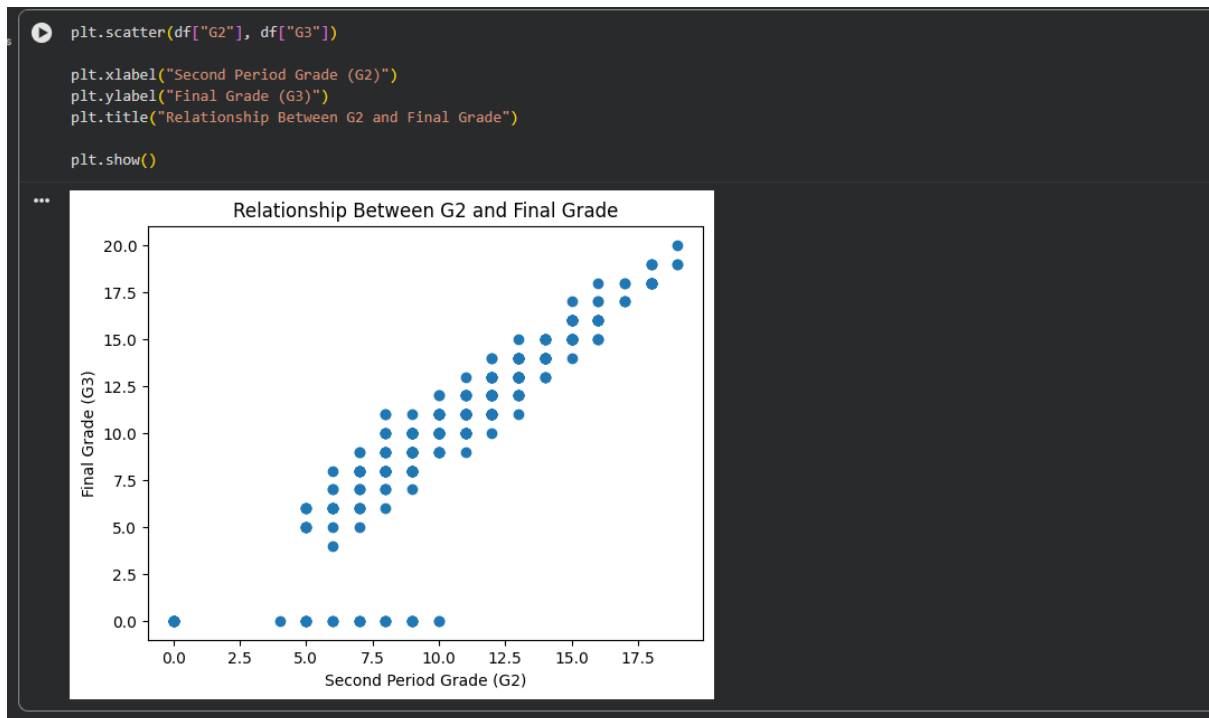


Figure 11: Relationship between second-period grade and final grade.

The scatter plot shows a clear positive relationship between G2 and G3. Students with higher second-period grades generally achieved higher final grades.

This supports the correlation result and suggests that **previous academic performance is a strong indicator of final academic performance.**

STUDENT PERFORMANCE CLASSIFICATION

To make the analysis more practically useful, students were divided into two performance categories based on their final grade:

- **Passing:** $G3 \geq 10$
- **At Risk:** $G3 < 10$

This classification allows the dataset to be viewed from an intervention perspective, where students below the selected threshold can be identified as potentially requiring additional academic support.

```
print("\nPerformance Group Percentage:")
print(
    df["performance_group"]
    .value_counts(normalize=True)
    .mul(100)
    .round(2)
)

...
Performance Group Percentage:
performance_group
Passing      67.09
At Risk     32.91
Name: proportion, dtype: float64
```

Figure 12: Distribution of students by performance group.

The classification resulted in:

Performance Group Students Percentage

Passing	265	67.09%
At Risk	130	32.91%

Therefore, approximately **one-third of the students were classified as At Risk** based on the selected threshold.

Characteristics of At-Risk and Passing Students

The two performance groups were compared across previous grades, final grade, failures, and absences.

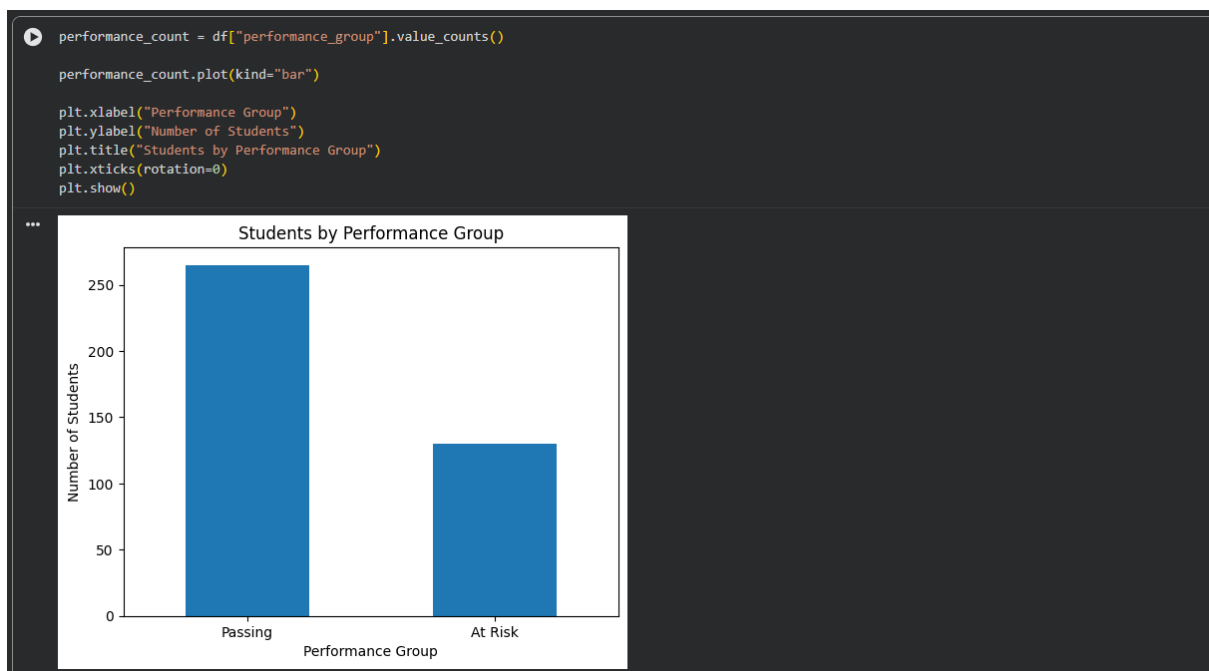


Figure 13: Comparative analysis of Passing and At-Risk students.

The comparison reveals a substantial difference between the two groups. Passing students had considerably higher average G1 and G2 scores and a much higher average final grade.

At-Risk students also had a higher average number of previous failures:

- **At Risk:** 0.69 failures
- **Passing:** 0.16 failures

This reinforces the earlier finding that previous academic performance and failure history are more useful indicators of academic risk than simply looking at attendance.

KEY FINDINGS

The major findings of the analysis are:

1. Previous academic grades are the strongest indicators

G2 had a correlation of approximately **0.905** with G3, while G1 had a correlation of approximately **0.802**.

2. Previous failures are negatively associated with performance

The correlation between failures and G3 was approximately **-0.360**. Students with more previous failures generally had lower final grades.

3. Study time has only a weak linear relationship

Study time had a correlation of approximately **0.098** with final grade. Although students in higher study-time categories generally performed somewhat better, the relationship was weak.

4. Absences were not a strong predictor

The correlation between absences and G3 was only approximately **0.034**, indicating almost no linear relationship in this dataset.

5. A significant proportion of students were classified as At Risk

Using a final-grade threshold of 10, **130 out of 395 students**, or approximately **32.91%**, were classified as At Risk.

6. The Passing group showed stronger academic history

Passing students had considerably higher G1 and G2 averages and fewer previous failures than the At-Risk group.

Practical Application

The analysis could be used as a basic academic early-warning system.

Instead of waiting until final examination results are available, an educational institution could use previous grades and failure history to identify students who may require additional support.

For example, students with:

- Low previous grades
- Multiple previous failures
- Low current academic performance

could be flagged for additional academic assistance, tutoring, mentoring, or closer monitoring.

The analysis also shows that relying on a single factor such as attendance or study time may not be sufficient. A combination of academic indicators provides a more meaningful picture of student performance.

Limitations of the Analysis

Although the analysis identifies several useful patterns, there are some limitations.

1. The dataset contains information from **395 students**, so the findings should not automatically be generalized to all students or educational institutions.
2. Correlation does not prove causation. For example, the strong relationship between G2 and G3 does not mean that G2 directly causes G3.
3. The analysis mainly focuses on relationships between variables and does not use advanced predictive machine-learning models.
4. The At-Risk classification uses a threshold of **G3 < 10**, which is a project-defined analytical rule and may differ from policies used by real educational institutions.
5. Some variables may interact with each other, meaning that a student's performance may depend on several factors simultaneously rather than one individual factor.

CONCLUSION

This project successfully demonstrated how SQL and Python can be combined to perform a practical student performance analysis.

SQL was used to perform structured data analysis through filtering, grouping, counting, and calculating averages. Python, using Pandas and Matplotlib, was then used to inspect the dataset, perform statistical analysis, identify relationships, and create visualizations.

The analysis of 395 students revealed several important patterns. The strongest relationship with final academic performance was found in previous academic grades, particularly G2, which had a correlation of approximately **0.905** with the final grade. G1 also showed a strong positive relationship with G3. In contrast, study time and absences showed very weak linear relationships with final grades.

Previous failures showed a negative relationship with final performance. Students with no previous failures achieved an average final grade of approximately **11.25**, while students with three previous failures achieved an average of approximately **5.69**. This indicates that previous academic difficulties may be an important indicator of future academic risk.

The project also classified students into Passing and At-Risk groups. **67.09% of students were classified as Passing, while 32.91% were classified as At Risk.** The At-Risk group had considerably lower previous and final grades and a higher average number of previous failures.

Overall, the project demonstrates that data analysis can transform a raw student dataset into useful information for decision-making. Rather than simply reporting grades, the analysis identifies patterns that could help educational institutions develop an early-warning approach and provide additional support to students who appear to be at greater academic risk.

The most important conclusion from the project is that **previous academic performance and failure history appear to provide stronger signals of final performance than study time or absence levels alone.** This highlights the value of using multiple data points when evaluating student performance and planning academic interventions.

RECOMMENDATIONS

Based on the findings, the following recommendations can be made:

1. **Monitor previous academic performance:** G1 and G2 can be used as important indicators when identifying students who may require support.
2. **Identify repeated academic difficulties:** Students with multiple previous failures should receive additional academic attention.
3. **Develop an early-warning system:** Schools could combine previous grades and failure history to identify potentially At-Risk students before final examinations.
4. **Avoid relying on attendance alone:** Since absences showed only a very weak relationship with G3, attendance should be considered alongside academic indicators rather than used as the sole measure of risk.
5. **Use data-driven academic support:** The results can help institutions target tutoring, mentoring, and other support resources toward students who need them most.

Tools and Technologies Used

Tool	Purpose
MySQL	SQL queries, grouping, aggregation and structured analysis
Python	Data analysis
Pandas	Data loading, cleaning and statistical analysis
Matplotlib	Data visualization
Google Colab	Python development environment
CSV Dataset	Source student-performance data

Project Outcome

Dataset: 395 students, 33 variables

Average Final Grade: 10.42

Highest Grade: 20

Lowest Grade: 0

Passing Students: 265 (67.09%)

At-Risk Students: 130 (32.91%)

Strongest Correlation: $G2 \rightarrow G3 = 0.905$

Failure Correlation: $\text{Failures} \rightarrow G3 = -0.360$

Study Time Correlation: 0.098

Absence Correlation: 0.034

References

1. UCI Machine Learning Repository. *Student Performance Dataset*.
2. Python Software Foundation. *Python Documentation*.
3. Pandas Documentation. *Pandas User Guide*.
4. Matplotlib Documentation. *Matplotlib Documentation*.
5. MySQL Documentation. *MySQL Reference Manual*.